Indoor Positioning via Wi-Fi Signal Strength Features and Classical Machine Learning

Team T076 — Indoor Wi-Fi Positioning Project
Department of Computer Science
University Project Report

***Abstract*—Accurate indoor positioning remains a critical challenge for location-based services where satellite-based solutions such as GPS are unreliable. This paper investigates Wi-Fi fingerprint-based indoor localization using the publicly available UJIIndoorLoc benchmark, which comprises over 21,000 received signal strength indicator (RSSI) scans from 520 wireless access points (WAPs) across three campus buildings and five floors. We present a systematic pipeline encompassing exploratory data analysis, principled preprocessing with sentinel-aware imputation and engineered aggregate features, and a comprehensive evaluation of classical machine learning baselines: k-Nearest Neighbours (KNN), Random Forest, Extra Trees, and XGBoost. An ablation study quantifies the contribution of each preprocessing stage. The best overall model, a Random Forest with 200 trees, achieves 99.7% building accuracy, 89.9% floor accuracy, and a median positioning error of 8.15 m on the benchmark validation set. These results show that well-engineered RSSI features combined with tree-based ensembles form a strong baseline for multi-building, multi-floor indoor localization.**

***Index Terms*—Indoor positioning, Wi-Fi fingerprinting, RSSI, Random Forest, XGBoost, k-nearest neighbours, UJIIndoorLoc**

I. Introduction

Indoor positioning is an essential enabler for navigation, asset tracking, emergency response, and context-aware services in environments where Global Navigation Satellite Systems (GNSS) cannot provide reliable coverage [1], [2]. Among the approaches explored, including Bluetooth Low Energy beacons, Ultra-Wideband ranging, and inertial dead reckoning [3], Wi-Fi fingerprinting remains the most practically attractive because it exploits existing wireless infrastructure without additional hardware deployment [4], [6].

The core idea of Wi-Fi fingerprinting is straightforward: the RSSI vector observed at a location forms a spatial "fingerprint" shaped by path loss, multi-path propagation, and structural attenuation [5]. A supervised learning model trained on a database of labelled fingerprints can then predict the location of a new observation.

A. Research Objectives

This work pursues the following objectives:

1) **Dataset understanding:** perform a rigorous exploratory analysis of the UJIIndoorLoc dataset [7] to quantify class balance, signal characteristics, and train–validation distributional shift.
2) **Feature engineering:** design a preprocessing pipeline that converts the raw 520-dimensional RSSI vector into a richer feature space through sentinel-aware imputation, binary missingness indicators, and engineered aggregate features (Section IV).
3) **Classical ML baselines:** implement and tune KNN [10], Random Forest [11], Extra Trees [12], and XGBoost [13] for joint building/floor classification and coordinate regression (Section V).
4) **Comprehensive evaluation:** compare all models on building accuracy, floor accuracy, joint accuracy, and mean, median, and 90th-percentile Euclidean positioning error, with building-wise and floor-wise breakdowns (Section VI).

B. Paper Organisation

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the UJIIndoorLoc dataset and our exploratory analysis. Section IV details the preprocessing and feature engineering pipeline. Section V presents the classical ML methodology and hyperparameter tuning. Section VI reports the experimental results. Section VII concludes the paper and outlines future directions.

II. Related Work

Wi-Fi fingerprint-based indoor positioning has been studied extensively since the seminal RADAR system [5]; recent surveys are given in [1], [2], [4]. Torres-Sospedra et al. [6] introduced the UJIIndoorLoc benchmark and reported baseline results using k-nearest neighbours. Their subsequent analysis [8] evaluated 51 combinations of distance and similarity measures and concluded that Manhattan-type distances with a small k were effective for fingerprinting.

Salamah et al. [9] applied machine learning techniques, including ensemble methods, to Wi-Fi indoor positioning and reported gains over basic KNN. Random Forests [11] and Extremely Randomized Trees [12] form the theoretical foundation for the tree-based ensembles evaluated here, while XGBoost [13] has become a default baseline for structured-data problems.

On the deep learning front, Nowicki and Wietrzykowski [15] proposed stacked autoencoders for dimensionality reduction followed by a DNN classifier, Kim et al. [16] designed a scalable DNN for multi-building, multi-floor localization, and Song et al. [17] applied convolutional networks to reshaped RSSI data. Multitask learning [18], [19] motivates jointly predicting building, floor, and coordinates in a shared architecture, and GradNorm [20] addresses

TABLE I
SUMMARY OF REPRESENTATIVE RELATED APPROACHES

Approach	Core idea	Ref.
RADAR	Signal-strength fingerprint matching by nearest neighbour	[5]
KNN variants	Systematic study of distance/similarity measures	[8]
Ensemble ML	Tree ensembles and boosting for positioning	[9]
Stacked AE + DNN	Unsupervised compression, then classification	[15]
Scalable DNN	Joint building/floor/location network	[16]
CNN	Convolutional features from RSSI maps	[17]

TABLE II
UJIINDOORLOC DATASET SUMMARY STATISTICS

Attribute	Train	Validation
Rows	19,937	1,111
Columns	529	529
Buildings	3	3
Floors	5	5
Building–floor pairs	13	13
Users	18	1
Phone models	16	11
% signal values = 100 (undetected)	96.54	96.83
Mean APs detected per scan	18.0	16.5
Exact duplicate rows	637	0

the task-balancing challenge. Table I summarises these approaches.

Our work contributes a thorough classical ML baseline with careful feature engineering and ablation analysis, establishing a performance reference that deep learning approaches should surpass to justify their additional complexity.

III. DATASET AND EXPLORATORY ANALYSIS

A. The UJIIndoorLoc Dataset

The UJIIndoorLoc dataset [6], [7] was collected at the Jaume I University campus in Castelló, Spain. It covers three multi-storey buildings with a total of 13 building–floor combinations. Each sample is a Wi-Fi fingerprint consisting of RSSI readings from 520 WAPs, together with metadata (user ID, phone ID, timestamp) and ground-truth labels (building ID, floor number, UTM longitude, and UTM latitude).

Table II summarises the key statistics of the training and validation splits. The training set comprises 19,937 fingerprints collected by 18 users with 16 phone models, while the validation set contains 1,111 fingerprints from a single user. This intentional design tests generalisation across users and across time.

B. Class Distribution

The classification targets exhibit moderate imbalance. Building 0 contains the most samples (9,116 training scans, 47.2% of the data), and the 13 building–floor classes have an imbalance ratio of 3.72:1 (largest to smallest class). Fig. 1 illustrates the joint building–floor class distribution.

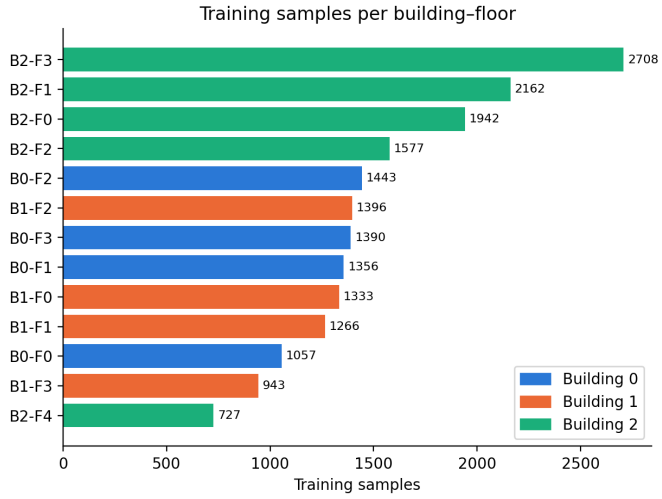


Fig. 1. Distribution of training samples across the 13 building–floor classes. Buildings 0 and 1 contain four floors each, and Building 2 contains five.

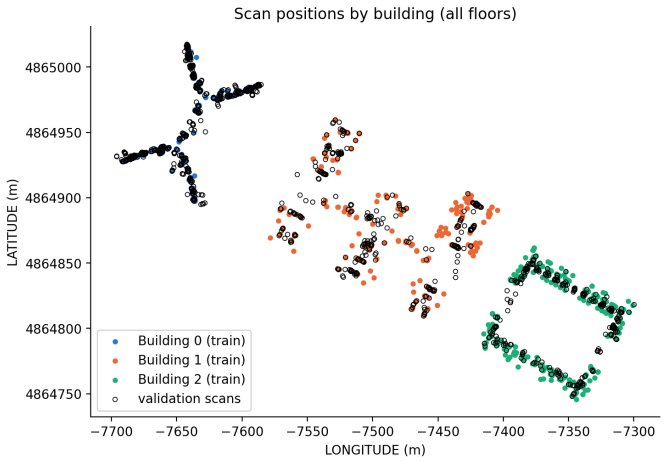


Fig. 2. Spatial distribution of Wi-Fi fingerprint collection points across the three campus buildings, in UTM coordinates.

C. Spatial Coverage

Fig. 2 shows the UTM coordinate coverage of each building. The three buildings are well separated in longitude, which explains the near-perfect building classification accuracy. Within each building, the coordinate spread motivates continuous regression.

D. RSSI Signal Characteristics

In UJIIndoorLoc, undetected access points are encoded with the sentinel value 100 rather than a null. The vast majority (96.5%) of the $520 \times 19{,}937$ RSSI matrix entries carry this sentinel, so each scan detects roughly 18 of the 520 WAPs on average. Detected signals span -104 to $0\,\text{dBm}$ with a mean of $-78.6\,\text{dBm}$ and a standard deviation of $12.7\,\text{dBm}$ (Fig. 3).

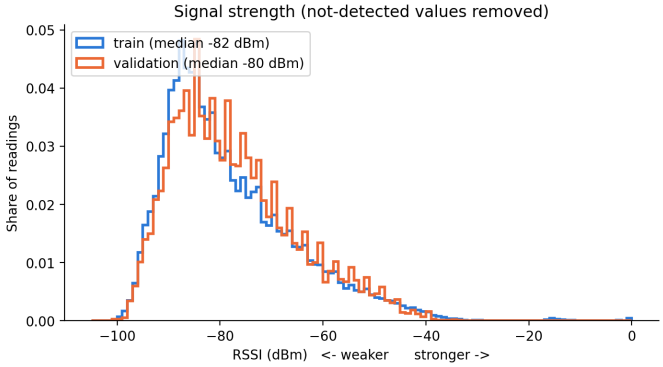


Fig. 3. Distribution of detected RSSI values (excluding the sentinel value 100) for the training and validation sets.

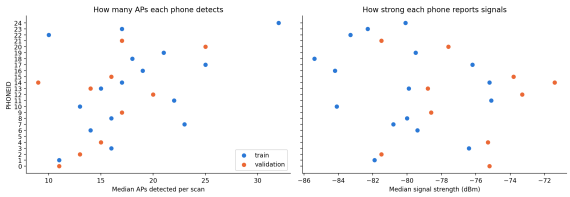


Fig. 4. Variation in detected APs and median signal strength across phone models.

E. Device Heterogeneity and Train–Validation Shift

The 16 phone models in the training set are highly heterogeneous: the median number of APs detected per scan ranges from 10 to 32, and the median signal strength varies by over 10 dBm across devices (Fig. 4). A Kolmogorov–Smirnov test confirms a statistically significant distributional shift between the training and validation sets ($p < 10^{-6}$) for both the number of APs per scan and the strongest signal per scan, underscoring the need for robust preprocessing.

IV. Feature Engineering and Preprocessing

The preprocessing pipeline consists of the stages shown in Fig. 5, each validated through the ablation experiments of Section IV-G.

A. Data Cleaning

We remove 637 exact duplicate rows from the training set and discard 76 scans with zero detected APs, as these carry no discriminative signal information.

B. Sentinel-Aware RSSI Imputation

The sentinel value 100 is replaced by -110 dBm, a value below the minimum observed RSSI (-104 dBm). This preserves the ordinal relationship among signal strengths while providing a consistent "very weak signal" level for tree-based models:

$$\mathrm{RSSI}'_i = \begin{cases} \mathrm{RSSI}_i & \text{if } \mathrm{RSSI}_i \neq 100, \\ -110 & \text{otherwise.} \end{cases} \tag{1}$$

TABLE III
Engineered Aggregate Features

Feature	Unit	Description
`num_aps`	count	Number of detected APs
`max_rssi`	dBm	Strongest detected signal
`mean_rssi`	dBm	Mean of detected signals
`std_rssi`	dBm	Std. dev. of detected signals
`range_rssi`	dBm	Max − min detected signal

TABLE IV
Ablation Study: Effect of Each Preprocessing Stage (Random Forest, 200 Trees)

Configuration	Bldg Acc	Floor Acc	Mean Err (m)
Raw (sentinel = 100)	99.9%	87.5%	13.98
+ Imputation (-110)	99.9%	91.2%	12.30
+ Indicators	99.8%	90.4%	12.21
+ Aggregates	99.8%	89.7%	12.82
+ Scaling (full)	99.8%	89.6%	12.80

C. Binary Missingness Indicators

For each of the 520 WAP columns, we create a binary indicator feature

$$m_i = \mathbb{I}(\mathrm{RSSI}_i = 100). \tag{2}$$

This provides the model with explicit information about the detection pattern, effectively encoding a "macro-fingerprint" that is highly discriminative for building and floor identification.

D. Engineered Aggregate Features

Five scan-level aggregate features capture the overall signal environment (Table III). Their distributions are visualised in Fig. 6.

E. Feature Scaling

All features are standardised to zero mean and unit variance using `StandardScaler` [14] fitted on the training set only, which prevents information leakage from the validation set.

F. Final Feature Space

After removing zero-variance columns, the feature vector comprises 465 imputed RSSI values, 465 binary missingness indicators, and 5 aggregate features, for a total of 935 features.

G. Ablation Study

Table IV and Fig. 7 present the ablation results. Replacing the sentinel value 100 by -110 dBm improves floor accuracy from 87.5% to 91.2% and reduces the mean positioning error from 13.98 m to 12.30 m. Adding missingness indicators and aggregate features yields only marginal changes, and scaling has a negligible effect for tree-based models, as expected since trees are invariant to monotonic feature transformations.

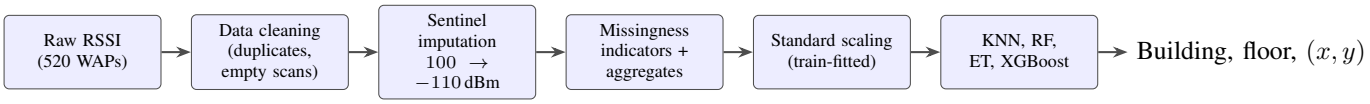


Fig. 5. Overview of the proposed preprocessing and prediction pipeline. Scaling parameters are fitted on the training set only.

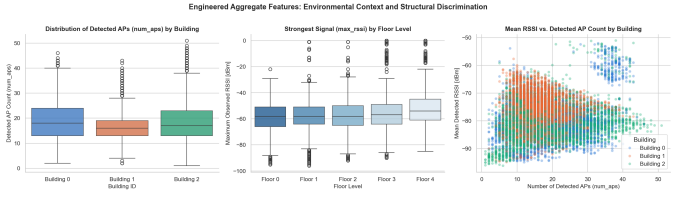


Fig. 6. Distributions of engineered aggregate features coloured by building, showing clear structural discrimination in `num_aps` and `mean_rssi`.

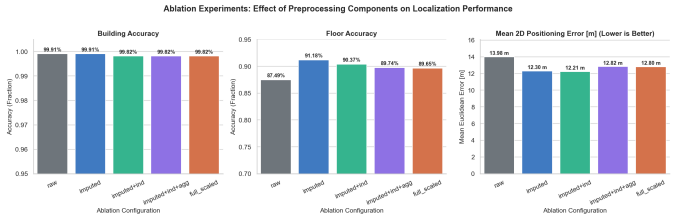


Fig. 7. Ablation study comparing five preprocessing configurations on building accuracy, floor accuracy, and mean positioning error.

TABLE V
KNN HYPERPARAMETER TUNING (5-FOLD CV ON TRAINING SET)

k	Weights	CV Error (m)	CV Floor Acc
1	uniform	10.75	90.1%
3	distance	9.91	89.7%
5	distance	9.67	89.6%
9	distance	9.51	89.6%
15	distance	9.50	89.5%

V. METHODOLOGY

We frame indoor positioning as three concurrent prediction tasks: (i) building classification (3 classes), (ii) floor classification (5 classes), and (iii) coordinate regression (UTM longitude and latitude in metres).

A. Models

1) Dummy Baseline: Predicts the most frequent building/floor and the median coordinates, establishing the minimum performance any useful model must exceed.

2) k-Nearest Neighbours (KNN): A distance-weighted KNN classifier/regressor [10]. Hyperparameters are tuned by 5-fold cross-validation on the training set (Table V).

3) Random Forest: An ensemble of decision trees trained on bootstrap samples with random feature subsets [11]. The search explores `n_estimators` $\in \{50, 100, 200\}$ and `min_samples_leaf` $\in \{1, 2, 5\}$ (Table VI). The best configuration (200 trees, `min_samples_leaf` = 1) achieves a cross-validated mean error of 9.65 m.

TABLE VI
RANDOM FOREST HYPERPARAMETER TUNING (5-FOLD CV)

Trees	Min Leaf	CV Error (m)	CV Floor Acc
50	1	9.75	97.2%
100	1	9.68	97.4%
200	1	**9.65**	**97.4%**
200	2	9.80	97.0%
200	5	10.41	96.1%

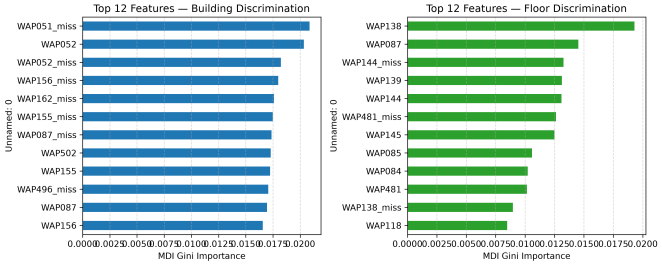


Fig. 8. Top features by Gini importance for the Random Forest building and floor classifiers.

4) Extra Trees: Similar to Random Forest but with randomised split thresholds [12], which adds diversity and often lowers variance.

5) XGBoost: A gradient-boosted tree ensemble [13]. Separate models are trained for building classification, floor classification, and coordinate regression.

B. Training Protocol

All models are trained on the 19,937-sample training set and evaluated on the 1,111-sample validation set, following the official UJIIndoorLoc protocol [6]. This simulates a realistic deployment in which the system is tested on data from an unseen user at a later time. All random seeds are fixed at 42 for reproducibility.

C. Feature Importance

Fig. 8 shows the top features ranked by Gini importance from the Random Forest building and floor classifiers. WAP RSSI features dominate, but the engineered aggregates (`num_aps`, `mean_rssi`) also rank highly, confirming their discriminative value.

VI. EXPERIMENTAL RESULTS

A. Evaluation Metrics

Models are evaluated on the following metrics.

- **Building and floor classification:** accuracy and macro-F1.

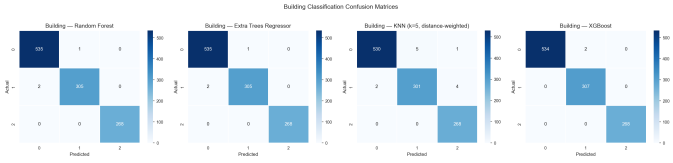


Fig. 9. Building classification confusion matrices for all models.

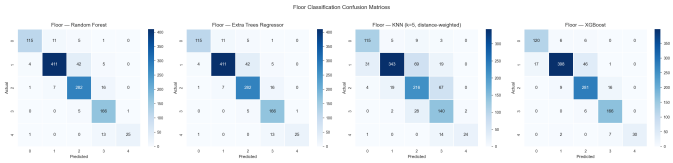


Fig. 10. Floor classification confusion matrices. Adjacent-floor confusion dominates the off-diagonal entries.

- **Joint building+floor accuracy:** both labels predicted correctly.
- **Coordinate regression:** mean absolute error (MAE), mean/median/90th-percentile Euclidean positioning error, RMSE, and R^2.

B. Overall Model Comparison

Table VII presents the comparison across all models. Random Forest and Extra Trees achieve identical classification performance (99.7% building, 89.9% floor), while Random Forest has the best overall balance with the highest R^2 (0.978). XGBoost excels at building classification (99.8%) but underperforms on coordinate regression (22.70 m mean error).

C. Classification Results

Table VIII reports detailed classification metrics. Building classification is near-perfect for all non-baseline models, reflecting the strong spatial separation between buildings. Floor classification is more challenging: Random Forest and Extra Trees reach 89.9% accuracy (macro-F1 = 0.874), whereas KNN drops to 75.4% owing to its sensitivity to the high-dimensional feature space. XGBoost achieves the best floor macro-F1 (0.892) despite slightly lower accuracy, indicating more balanced per-floor performance.

D. Confusion Matrices

Figs. 9 and 10 show the confusion matrices for building and floor classification. Building confusion is minimal (1–3 misclassifications for the best models). Floor confusion is concentrated between adjacent floors, particularly Floors 1–2 and 2–3, consistent with the physics of inter-floor signal attenuation.

E. Coordinate Regression Results

Table IX presents the coordinate regression metrics. Random Forest achieves the lowest MAE (7.92 m) and mean error, while Extra Trees obtains the lowest median error (7.66 m) and 90th-percentile error (25.68 m), indicating a tighter error distribution with fewer outliers.

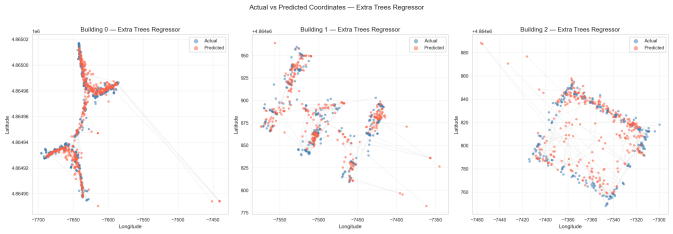


Fig. 11. Actual versus predicted UTM coordinates for Random Forest and Extra Trees.

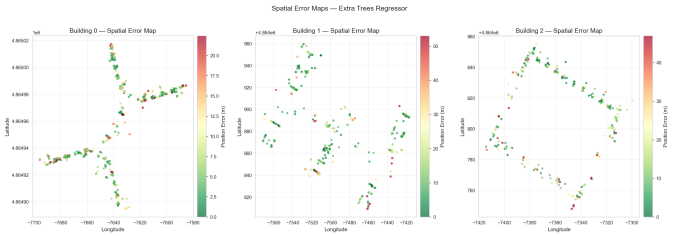


Fig. 12. Spatial distribution of Euclidean positioning errors, coloured by error magnitude.

F. Actual vs. Predicted Coordinates

Fig. 11 plots actual versus predicted coordinates for the best-performing models. Predictions closely track the ground truth, with larger errors at building boundaries and in sparsely sampled areas.

G. Spatial Error Analysis

Fig. 12 visualises the spatial distribution of positioning errors. Errors are lowest in Building 0 (mean 8.48 m for Random Forest), where training data density is highest, and largest in Building 2 (mean 17.46 m).

H. Building-Wise Error Breakdown

Table X breaks down Random Forest performance by building. Building 0 achieves the best results on all metrics, consistent with its larger training sample size and denser spatial coverage.

I. Error Distribution and CDF

Fig. 13 shows the positioning error distribution for all models. The Random Forest and Extra Trees distributions are heavily right-skewed, with most errors below 15 m and a long tail extending to 30–40 m.

Fig. 14 presents the cumulative distribution function (CDF) of positioning errors. At the 10 m threshold, Random Forest resolves 62% of samples and Extra Trees 64%.

J. Model Comparison Summary

Fig. 15 summarises all models across key metrics. Random Forest and Extra Trees dominate on both classification and regression, while XGBoost excels at classification but struggles with coordinate regression.

TABLE VII
COMPREHENSIVE MODEL COMPARISON ON THE UJIINDOORLOC VALIDATION SET

Model	Bldg Acc	Bldg F1	Floor Acc	Floor F1	Joint Acc	Mean Err (m)	Med Err (m)	P90 Err (m)	R^2
Random Forest	99.7%	0.997	89.9%	0.874	89.7%	12.30	8.15	29.55	0.978
Extra Trees	99.7%	0.997	89.9%	0.874	89.7%	12.46	**7.66**	**25.68**	0.971
XGBoost	**99.8%**	**0.998**	89.6%	**0.892**	89.6%	22.70	15.17	47.77	0.940
KNN (k=5, dist.)	98.9%	0.988	75.4%	0.748	75.2%	13.68	8.20	29.21	0.964
Dummy Baseline	24.1%	0.130	15.5%	0.054	3.6%	161.19	142.34	263.48	−0.634

TABLE VIII
DETAILED CLASSIFICATION RESULTS (%, F1 AS FRACTION)

Model	Target	Acc	Prec	Rec	F1
Random Forest	Building	99.7	99.8	99.7	0.997
XGBoost	Building	99.8	99.8	99.9	0.998
KNN	Building	98.9	98.7	99.0	0.988
Random Forest	Floor	89.9	90.8	85.8	0.874
XGBoost	Floor	89.6	90.8	88.5	0.892
KNN	Floor	75.4	77.2	75.0	0.748

TABLE IX
COORDINATE REGRESSION RESULTS

Model	MAE (m)	Mean Err (m)	Med Err (m)	P90 Err (m)
Random Forest	7.92	12.30	8.15	29.55
Extra Trees	8.01	12.46	**7.66**	**25.68**
KNN	8.82	13.68	8.20	29.21
XGBoost	14.70	22.70	15.17	47.77

TABLE X
RANDOM FOREST PERFORMANCE BREAKDOWN BY BUILDING

Building	Floor Acc	Mean (m)	Median (m)	P90 (m)
Building 0	96.8%	8.48	5.31	18.82
Building 1	80.5%	14.47	10.12	35.69
Building 2	86.9%	17.46	11.56	36.95

K. Floor-Wise Error Analysis

Fig. 16 presents the positioning error by floor. Floors 0 and 3 tend to have higher errors, possibly owing to edge effects and proximity to building entrances where signal conditions change rapidly.

L. Failure Case Analysis

Fig. 17 maps the worst-case predictions (top 10% of errors). Large errors concentrate at building boundaries and along corridors where the fingerprint changes rapidly over short distances.

VII. CONCLUSION AND FUTURE WORK

This paper presented a systematic investigation of Wi-Fi fingerprint-based indoor positioning on the UJIIndoorLoc benchmark. The principal findings are as follows.

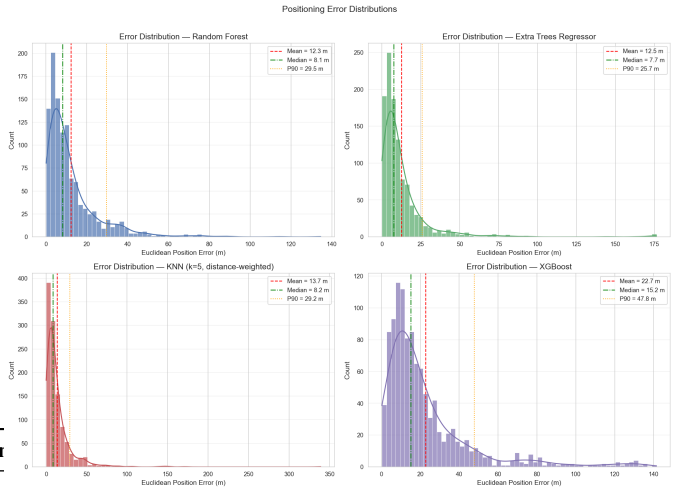


Fig. 13. Positioning error distribution (histogram and density) for each model.

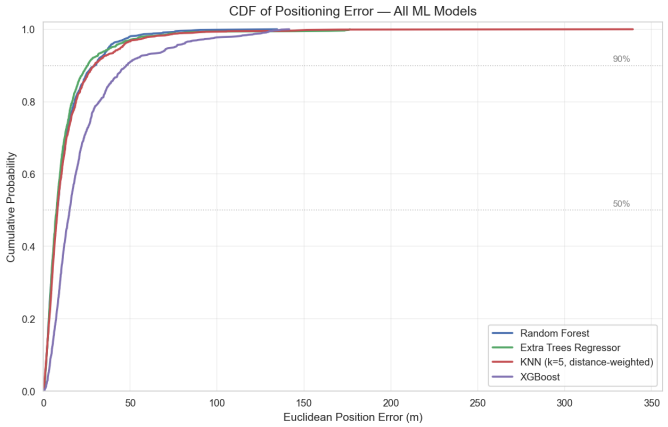


Fig. 14. Cumulative distribution of Euclidean positioning error. Steeper curves indicate better localisation accuracy.

1) **Preprocessing matters:** sentinel-aware imputation (100 → −110 dBm) alone improves floor accuracy by 3.7 percentage points and reduces the mean positioning error by 1.68 m.
2) **Classical ML is competitive:** Random Forest with 200 trees achieves 99.7% building accuracy, 89.9% floor accuracy, and 8.15 m median positioning error, a strong reference point for deep learning approaches [15]–[17].

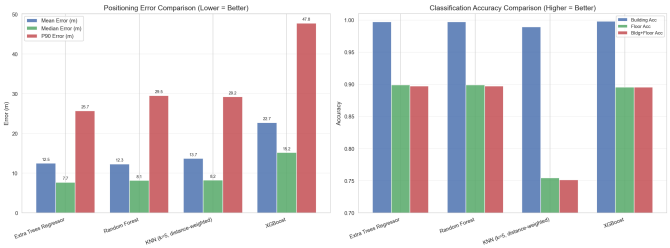


Fig. 15. Multi-metric model comparison summarising building accuracy, floor accuracy, and positioning error.

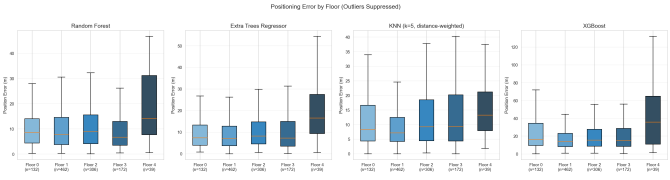


Fig. 16. Mean positioning error by floor for each model.

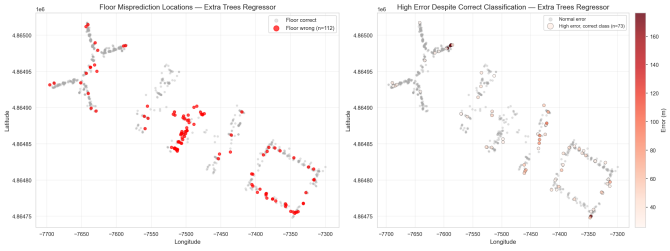


Fig. 17. Spatial distribution of failure cases (top 10% positioning errors). Arrows connect predicted to actual locations.

3) **Extra Trees gives tighter error bounds:** with identical classification accuracy, it obtains the lowest median (7.66 m) and 90th-percentile (25.68 m) errors.
4) **XGBoost excels at classification, not regression:** it attains the best building F1 (0.998) and floor F1 (0.892) but roughly twice the positioning error of Random Forest.
5) **Building-dependent performance:** all models perform best in Building 0 (largest training set) and worst in Buildings 1 and 2, suggesting that data augmentation or building-specific models could help.

A. Future Directions

Several directions remain open:

- **Deep multitask learning:** a shared DNN with hard parameter sharing [18], [19] could exploit cross-task regularisation.
- **Stacked autoencoders:** unsupervised pretraining to learn a compressed RSSI representation [15].
- **Adaptive loss balancing:** techniques such as GradNorm [20] could balance classification and regression losses automatically.
- **Data augmentation:** synthetic fingerprint generation through signal perturbation or generative models to address under-represented building–floor pairs.
- **Online adaptation:** incremental model updates as new crowdsourced fingerprints become available.

References

[1] X. Feng, K. A. Nguyen, and Z. Luo, "A survey of deep learning approaches for WiFi-based indoor positioning," *J. Inf. Telecommun.*, vol. 6, no. 2, pp. 163–216, 2022.

[2] F. Zafari, A. Gkelias, and K. K. Leung, "A survey of indoor localization systems and technologies," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 3, pp. 2568–2599, 3rd Quart., 2019.

[3] P. Davidson and R. Piché, "A survey of selected indoor positioning methods for smartphones," *IEEE Commun. Surveys Tuts.*, vol. 19, no. 2, pp. 1347–1370, 2nd Quart., 2017.

[4] S. He and S.-H. G. Chan, "Wi-Fi fingerprint-based indoor positioning: Recent advances and comparisons," *IEEE Commun. Surveys Tuts.*, vol. 18, no. 1, pp. 466–490, 1st Quart., 2016.

[5] P. Bahl and V. N. Padmanabhan, "RADAR: An in-building RF-based user location and tracking system," in *Proc. IEEE INFOCOM*, Tel Aviv, Israel, Mar. 2000, pp. 775–784.

[6] J. Torres-Sospedra *et al.*, "UJIIndoorLoc: A new multi-building and multi-floor database for WLAN fingerprint-based indoor localization problems," in *Proc. Int. Conf. Indoor Positioning Indoor Navigat. (IPIN)*, Busan, South Korea, Oct. 2014, pp. 261–270.

[7] J. Torres-Sospedra *et al.*, "UJIIndoorLoc," UCI Machine Learning Repository, 2014. [Online]. Available: https://archive.ics.uci.edu/dataset/310/ujiindoorloc

[8] J. Torres-Sospedra, R. Montoliu, S. Trilles, Ó. Belmonte, and J. Huerta, "Comprehensive analysis of distance and similarity measures for Wi-Fi fingerprinting indoor positioning systems," *Expert Syst. Appl.*, vol. 42, no. 23, pp. 9263–9278, Dec. 2015.

[9] A. H. Salamah, M. Tamazin, M. A. Sharkas, and M. Khedr, "An enhanced WiFi indoor localization system based on machine learning," in *Proc. Int. Conf. Indoor Positioning Indoor Navigat. (IPIN)*, Alcalá de Henares, Spain, Oct. 2016, pp. 1–8.

[10] T. Cover and P. Hart, "Nearest neighbor pattern classification," *IEEE Trans. Inf. Theory*, vol. IT-13, no. 1, pp. 21–27, Jan. 1967.

[11] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, Oct. 2001.

[12] P. Geurts, D. Ernst, and L. Wehenkel, "Extremely randomized trees," *Mach. Learn.*, vol. 63, no. 1, pp. 3–42, Apr. 2006.

[13] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, San Francisco, CA, USA, Aug. 2016, pp. 785–794.

[14] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, Oct. 2011.

[15] M. Nowicki and J. Wietrzykowski, "Low-effort place recognition with WiFi fingerprints using deep learning," in *Automation 2017*, Adv. Intell. Syst. Comput., vol. 550. Cham, Switzerland: Springer, 2017, pp. 575–584.

[16] K. S. Kim, S. Lee, and K. Huang, "A scalable deep neural network architecture for multi-building and multi-floor indoor localization based on Wi-Fi fingerprinting," *Big Data Anal.*, vol. 3, no. 1, p. 4, Apr. 2018.

[17] X. Song *et al.*, "A novel convolutional neural network based indoor localization framework with WiFi fingerprinting," *IEEE Access*, vol. 7, pp. 110698–110709, 2019.

[18] R. Caruana, "Multitask learning," *Mach. Learn.*, vol. 28, no. 1, pp. 41–75, Jul. 1997.

[19] S. Ruder, "An overview of multi-task learning in deep neural networks," *arXiv:1706.05098*, 2017.

[20] Z. Chen, V. Badrinarayanan, C.-Y. Lee, and A. Rabinovich, "GradNorm: Gradient normalization for adaptive loss balancing in deep multitask networks," in *Proc. 35th Int. Conf. Mach. Learn. (ICML)*, Stockholm, Sweden, Jul. 2018, pp. 794–803.